

Length of surreal product

OPG-37416

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Conjecture Every surreal number has a unique sign expansion, i.e. function $f : o \rightarrow \{-, +\}$, where o is some ordinal. This o is the length of given sign expansion and also the birthday of the corresponding surreal number. Let us denote this length of s as $\ell(s)$.

It is easy to prove that

$$\ell(s + t) \leq \ell(s) + \ell(t)$$

What about

$$\ell(s \times t) \leq \ell(s) \times \ell(t)$$

?